

Initial configuration

```
TRAINING_CONFIG = {
    "num_episodes": 3000,
    "timesteps_per_episode": 50,
    "T_horizon": 64,
    "sma_window_size": 100,
    "random" : True,
    "hero_starting_pos": [1, 1],
    "villain_starting_pos": [7, 7],
    "step_penalty": -0.1,
    "win_reward": +5,
    "lose_reward": -5,
}

PPO_CONFIG_Hero = {
    "state_dim": 4,
    "action_dim": 4,
    "hidden_dim": 256,
    "lr": 0.0005,
    "gamma": 0.98,
    "lmbda": 0.95,
    "eps_clip": 0.1,
    "K_epoch": 3,
}

PPO_CONFIG_Villan = {
    "state_dim": 4,
    "action_dim": 4,
    "hidden_dim": 256,
    "lr": 0.0005,
    "gamma": 0.98,
    "lmbda": 0.95,
    "eps_clip": 0.1,
    "K_epoch": 3,
}
```

Keep the above configuration as reference, and only change the below given parameters.

Do not forget to add experiment number while running the code.

Make the changes in ppo_config file. And then run the train_ppo file

Exp 1 : Hero_lr=0.001, Villan_lr=0.001
Exp 2 : Hero_lr=0.0005, Villan_lr=0.0005
Exp 3 : Hero_lr=0.0001, Villan_lr=0.0001
Exp 4 : Hero_lr=0.00005, Villan_lr=0.00005

Exp 5 : Hero_lr=0.001, Villan_lr=0.0005
Exp 6 : Hero_lr=0.001, Villan_lr=0.0001
Exp 7 : Hero_lr=0.001, Villan_lr=0.00005

Exp 8 : Hero_lr=0.0005, Villan_lr=0.001
Exp 9 : Hero_lr=0.0005, Villan_lr=0.0001
Exp 10 : Hero_lr=0.0005, Villan_lr=0.00005

Exp 11 : Hero_lr=0.0001, Villan_lr=0.001
Exp 12 : Hero_lr=0.0001, Villan_lr=0.0005
Exp 13 : Hero_lr=0.0001, Villan_lr=0.00005

Exp 14 : Hero_lr=0.00005, Villan_lr=0.001
Exp 15 : Hero_lr=0.00005, Villan_lr=0.0005
Exp 16 : Hero_lr=0.00005, Villan_lr=0.0001

Exp 17 : Hero_gamma=0.99, Villan_gamma=0.99
Exp 18 : Hero_gamma=0.99, Villan_gamma=0.95
Exp 19 : Hero_gamma=0.99, Villan_gamma=0.90

Exp 20 : Hero_gamma=0.95, Villan_gamma=0.99
Exp 21 : Hero_gamma=0.95, Villan_gamma=0.95
Exp 22 : Hero_gamma=0.95, Villan_gamma=0.90

Exp 23 : Hero_gamma=0.90, Villan_gamma=0.99
Exp 24 : Hero_gamma=0.90, Villan_gamma=0.95
Exp 25 : Hero_gamma=0.90, Villan_gamma=0.90

Exp 26 : Hero_lambda=0.99, Villan_lambda=0.99
Exp 27 : Hero_lambda=0.99, Villan_lambda=0.95
Exp 28 : Hero_lambda=0.99, Villan_lambda=0.90

Exp 29 : Hero_lambda=0.95, Villan_lambda=0.99

Exp 30 : Hero_lambda=0.95, Villan_lambda=0.95

Exp 31 : Hero_lambda=0.95, Villan_lambda=0.90

Exp 32 : Hero_lambda=0.90, Villan_lambda=0.99

Exp 33 : Hero_lambda=0.90, Villan_lambda=0.95

Exp 34 : Hero_lambda=0.90, Villan_lambda=0.90

Exp 35 : Hero_eps_clip=0.1, Villan_eps_clip=0.1

Exp 36 : Hero_eps_clip=0.1, Villan_eps_clip=0.2

Exp 37 : Hero_eps_clip=0.1, Villan_eps_clip=0.3

Exp 38 : Hero_eps_clip=0.2, Villan_eps_clip=0.1

Exp 39 : Hero_eps_clip=0.2, Villan_eps_clip=0.2

Exp 40 : Hero_eps_clip=0.2, Villan_eps_clip=0.3

Exp 41 : Hero_eps_clip=0.3, Villan_eps_clip=0.1

Exp 42 : Hero_eps_clip=0.3, Villan_eps_clip=0.2

Exp 43 : Hero_eps_clip=0.3, Villan_eps_clip=0.3

Exp 44 : Hero_k_epoch=3, Villan_k_epoch=3

Exp 45 : Hero_k_epoch=3, Villan_k_epoch=5

Exp 46 : Hero_k_epoch=3, Villan_k_epoch=9

Exp 47 : Hero_k_epoch=5, Villan_k_epoch=3

Exp 48 : Hero_k_epoch=5, Villan_k_epoch=5

Exp 49 : Hero_k_epoch=5, Villan_k_epoch=9

Exp 50 : Hero_k_epoch=9, Villan_k_epoch=3

Exp 51 : Hero_k_epoch=9, Villan_k_epoch=5

Exp 52 : Hero_k_epoch=9, Villan_k_epoch=9

Exp 53 : Hero_T_horizon=64

Exp 54 : Hero_T_horizon=128

Exp 55 : Hero_T_horizon=256

Exp 56 : Hero_gamma=0.85, Villan_gamma=0.99

Exp 57 : Hero_gamma=0.99, Villan_gamma=0.85

Exp 58 : Hero_gamma=0.80, Villan_gamma=0.99

Exp 59 : Hero_gamma=0.99, Villan_gamma=0.80

Exp 60 : Hero_gamma=0.85, Villan_gamma=0.80

Exp 61 : Hero_gamma=0.80, Villan_gamma=0.85

Exp 62 : step_penalty= 0

Exp 63 : step_penalty= -0.1

Exp 64 : step_penalty= -0.05

Exp 65 : step_penalty= -0.01

Exp 66 : Hero_hidden_dim=128, Villan_hidden_dim =128

Exp 67 : Hero_hidden_dim=256, Villan_hidden_dim =256

Exp 68 : timesteps_per_episode = 25

Exp 69 : timesteps_per_episode = 50

Exp 70 : timesteps_per_episode = 75

Exp 71 : timesteps_per_episode = 100